GLOBAL ENERGY HACKATHON 2026 · BRIEF 03 · DIGITAL SOLUTIONS & AI

What P90 did for solar finance, Lodestar does for mini-grid demand.

Developers must size a mini-grid before a single meter exists. Today they guess, and lenders can't price a guess.

217,000
mini-grids needed to serve 490 million people by 2030, $127B (ESMAP)

65%
error of survey-based demand estimates per customer (CrossBoundary Innovation Lab)

12.5% → 6.1%
IRR of a 30 kW site when oversized by 100% (CrossBoundary)

None
of the open tools we found combine a demand *range*, a *growth* curve and a lender view, validated against meters

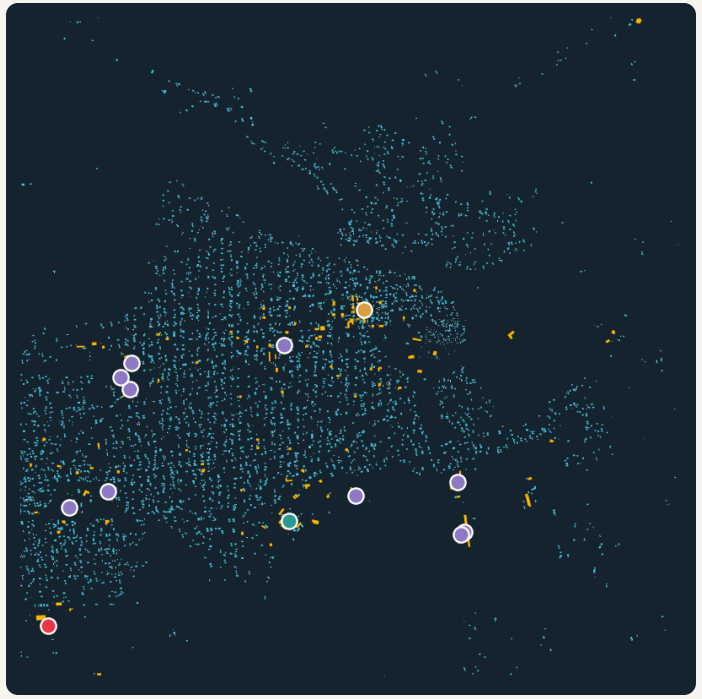
Who: mini-grid developers planning sites in unelectrified villages (Nigeria, Kenya, DRC first). **Who pays:** the lenders, DFIs and results-based-financing programmes that fund them. **Who benefits:** households and businesses that get a system sized to them, at a tariff they can afford, without evening brownouts.

Team Lodestar · Tushar Agarwal

lodestar-utfz.onrender.com · github.com/TusharTechs/lodestar

THE SOLUTION · BUILT THIS WEEKEND

Click a village. Get a bankable demand forecast.

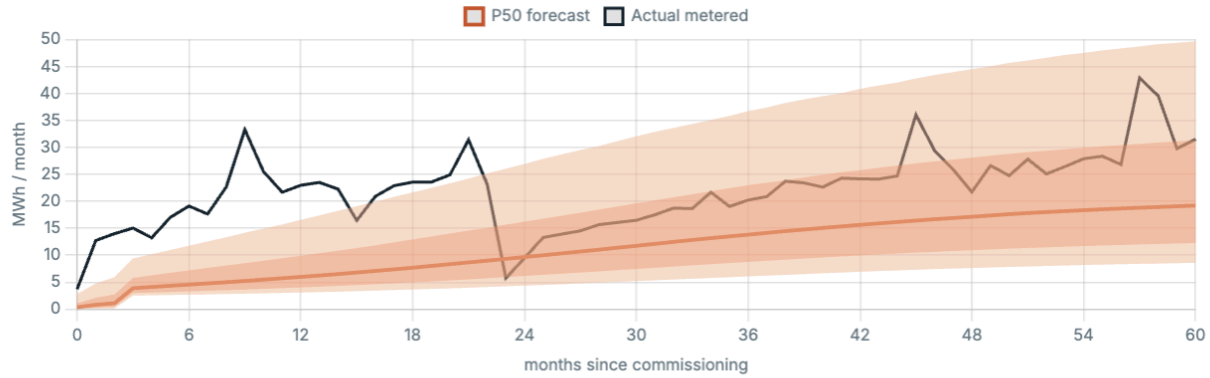


Mutwanga (DRC): 11,400 open building footprints → ~3,800 potential households, 358 businesses; schools, clinic, market, mill from OpenStreetMap.

YEAR-3 DEMAND · P50	P90 · CONSERVATIVE	P10 · OPTIMISTIC	PEAK LOAD · P50
11.9 MWh/mo	**5.3**	**32.4**	**35 kW**
Confidence: **low** · P10/P90 = 6.1×	exceeded with 90% probability	exceeded with 10% probability	load factor 39%

Demand growth, months 0–60 after commissioning

Monthly site energy (MWh). Dark band P75–P25, light band P90–P10. **Black line: actual metered demand**, which this forecast did not see.



Demand Bureau: share the first 6 metered months →

re-weights the forecast with the operator's own meters

Year-3 demand 11.9 MWh/month (P90 5.3, P10 32.4), forecast from footprints and OpenStreetMap alone; it learns only from the other metered sites.

The black line is the site's **real metered demand**, which the model never saw.

Below in the report: driver waterfall, load profile, sizing options, IRR distribution and cash flow at P90/P50/P10.

Range, not a point: P90/P50/P10 from 20,000 Monte Carlo draws

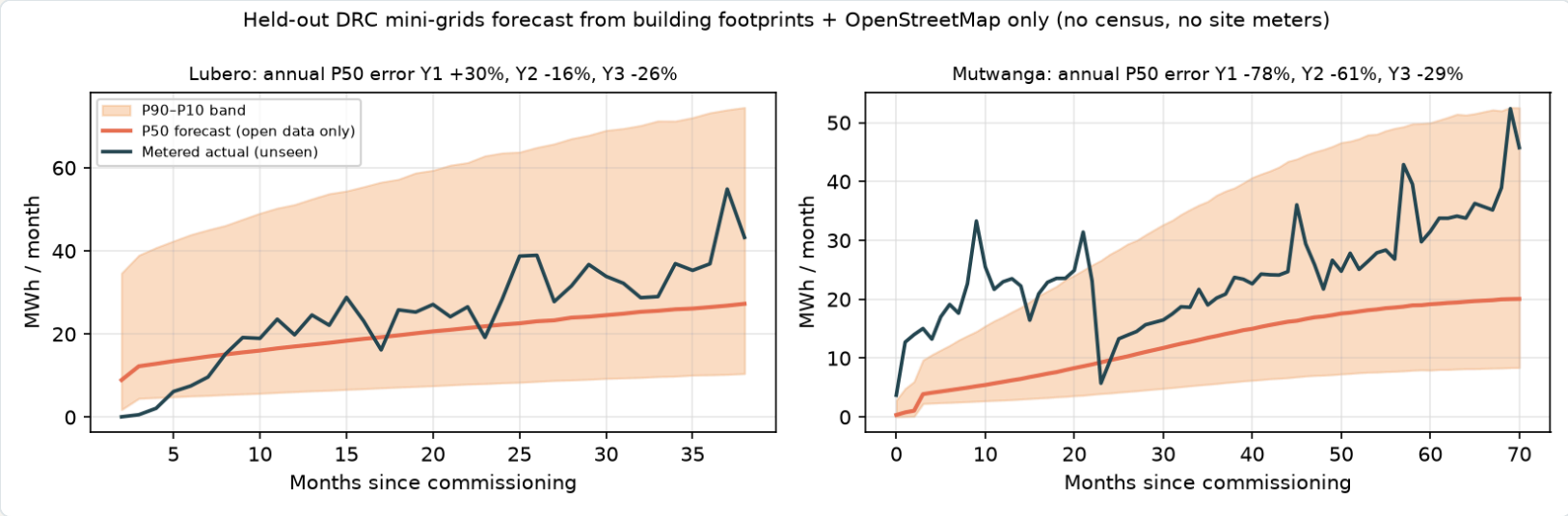
Growth: connections × kWh per connection, months 0–60

Explained in plain words: what each forecast is built from, with every source dated and every driver tagged

Lender view: sizing, grant needed, debt safe on P90, memo, HOMER 8760

PROOF · THE BRIEF'S METRIC: ERROR VS METERED DEMAND ON HELD-OUT SITES

Validated against real meters, using open data alone.



No partner meter data was available, so we found a public CC0 panel: three metered mini-grids in North Kivu, DRC, 2019–2024 (Lunanga et al. 2026). Each village site is forecast with **no census and none of its own meters**; calibration ratios and growth shapes come from the *other* sites only.

5 of 6
site-years of metered energy inside the P90–P10 band (target 80%)

81%
of metered months inside the band

−26% / −29%
year-3 (sizing-year) error, Lubero / Mutwanga · ref. 37% per-site ML (Allee et al.)

Honest limits, all shown in the app: two village sites is method validation; a census-based flat estimate scored 36% on them but gives no range; after 12 metered months, shocks (Mutwanga's conflict collapse) make persistence the better predictor.

Reproduce: python scripts/validate_drc.py

FEASIBILITY, COST & WHO PAYS

A trust layer the sector can afford, and that improves as it's used.

Who pays

- **Free** single-site forecasts (adoption)
- **Bankable Demand Report** per site for developers and lenders (proposed ~$300, under 0.1% of a ~$585k mini-grid)
- **Portfolio screening** subscriptions for DFIs and RBF programmes
- Households pay nothing; each report checks their bill against the AMDA ARPU benchmark

Measured: 0.76 CPU-s and 6 KB per forecast. **Break-even:** ~200 paid reports/yr, 3.3% of the 6,116 sites AMDA members plan (team cost assumed $60k/yr). Open data, no API keys, runs on a laptop.

Demand Bureau: safe meter-data sharing

- **Give-to-get**, borrowed from credit bureaus: share site-month aggregates, get benchmarks and narrower bands
- No customer data; release only cells with ≥3 sites from ≥2 operators
- Federated option: share fitted parameters, not series
- Neutral steward (e.g. AMDA); a pooling clause in RBF grants seeds it

Forecast ledger

Borrowed from weather-forecast verification and clinical-trial registries: every forecast is **hashed and timestamped before the site is built**, then scored publicly when meters report. Trust earned in the open, not claimed.

6–18 months

M0–3
Backtest 2–3 operators' historical meters under the Bureau protocol

M3–6
Pre-register forecasts for 10 sites before commissioning

M6–12
Score them publicly; integrate with one RBF programme's site validation

M12–18
5 countries; paid reports to lenders · DTECH 2027 MVP milestone

Biggest risk: thin metered data. Tested early by the M0–3 operator backtest; mitigated by the Bureau and the ledger.

IMPACT & SCALE · SAME REAL METERS, TURNED INTO DECISIONS

Right-sized, financeable, and reachable from any phone.

−72%
capacity mismatch at year 3 vs a desk-estimate build, replaying real metered demand (155 → 44 kWp, two DRC sites)

+37% → +6%
Mutwanga: desk build oversized; Lodestar's build-at-P50-then-expand plan was within 6%

−49%
cost of power when sized on P50 instead of the optimistic P10 (avg. of 20 villages)

$0.20–0.40
/kWh tariff needed by mini-grid candidates with a Nigeria-NEP-style $600/connection grant, vs ~$1 unsubsidised

Portfolio screener for RBF programmes

20 villages in 15 CPU-seconds: **1** bankable with an RBF grant, **4** need an anchor load, **12** are solar-home-system territory, **3** near the grid. Telling a programme where *not* to build is impact too. Nigeria's DARES alone targets 1,000+ mini-grids.

Built for the field, and for anywhere with rooftops

USSD + SMS: a forecast in one 160-character text on a feature phone. **Offline pack:** 12 KB, opens without internet. Footprints and OpenStreetMap cover all of Africa, South/Southeast Asia and Latin America; a new country needs only its regional benchmark.

Lodestar makes demand bankable, so the next 200,000 mini-grids get built at the right size.

Next: 2–3 operators for the M0–3 backtest, one RBF programme as pilot partner, a neutral host for the Demand Bureau.